



Amlogic Hybrid Android S Release Note

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Revision History

V1.1

Update xts/nts/license version info;
Mandatory Patch(->2023/05/25);
Security Patch(->2023/05/05);
Update the CBS configure version;

V1.0

2023/04/15 Release(AB4->RC1);
Mandatory Patch(->2023/02/28);
Security Patch(->2023/03/05);

Amlogic Openlinux Release Notes

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Release Information

1.1. Release Milestone Introduction

This document is the release notes of Amlogic reference source codes aligned with Android S running on Amlogic S905X4/S905C2/S905C2L/S905Y4/S905C3/S805X2 reference hardware.

S905X4/S905C2/S905C2L/S905Y4/S905C3/S805X2 released the GTVS clean version(Hybrid RC1).

The purpose of this release is:

1. GTVS Clean.
 - (1) CTS(12.0 R8)
 - (2) GTS(10.0 R4)
 - (3) VTS(12.0 R8)
 - (4) CTS-verify(12.0 R7)
 - (5) CTS-on-gsi(12.0 R8)
 - (6) STS(2023-05)
 - (7) TVTS(2.6 R1)
 - (8) Android TV smoking
2. NTS pre-cert(ninja 9)
3. Pre-certification
 - (1) VMX-IPTV Client E2E
 - (2) VMX-WEB Client E2E
 - (3) VMX-DVB Client E2E(DTVKit)
 - (4) NAGRA TKL Simbad for S905X4(DTVKit)
 - (5) NAGRA NOCS Simbad for S905C2/S905C2L(DTVKit)
 - (6) NAGRA TKL for S905X4(CBS)
 - (7) NAGRA NOCS for S905C2(CBS)
 - (8) Irdeto for S905C2/S905C3(Irdeto middleware)
 - (9) Irdeto for S905C3(DTVKit)
 - (10) NAGRA TKL Simbad (CBS)
 - (11) NAGRA NOCS Simbad (CBS)
 - (12) Widevine CAS CBS
 - (13) DDP V4.8 for DVB
 - (14) MS12 V2 for DVB
 - (15) MS12 V1 DVB for S905X4
 - (16) Dolby vision v2.6
 - (17) DTS-HD
 - (18) ART
 - (19) HDMI 2.0
 - (20) HDCP1.4
 - (21) HDCP2.3
4. Full function ready:
 - (1) Component compatibility
 - (1) HDMI
 - (2) CVBS
 - (3) WIFI
 - (4) BT
 - (5) Ethernet
 - (6) Bluetooth
 - (7) USB
 - (8) Remote
 - (9) Spdif
 - (2) Suspend/Resume

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- (3) Multi-media functionality.
- (4) Multi-instance.
- (5) Widevine CAS.
- (6) Display functionality.
- (7) PQ/DI.
- (8) HDR10.
- (9) Dolby Vision.
- (10) System functionality.
- (11) Audio-output.
- (12) Secure boot V3.
- (13) Fastboot
- (14) A/B update
- (15) AVB Anti-roll back
- (16) HDMI CTS 2.0
- (17) HDMI CEC
- (18) HDR10+
- (19) HDCP 1.4/2.3
- (20) CBS
- (21) Synamedia
- (22) USB-CI
- (23) Sound Bar

5. Performance

- (1) Benchmark
- (2) UX
- (3) APP launch speed
- (4) Video start/Switch speed
- (5) Memory usage
- (6) Boot Time

6. DTVKit

- (1) DVB-C Program search
- (2) DVB-S/DVB-S2 Program search
- (3) DVB-T/DVB-T2 Program search
- (4) Channel auto search
- (5) Channel manual search
- (6) NIT search
- (7) MHEG5
- (8) Teletext
- (9) PVR
- (10) TimeShift
- (11) subtitle
- (12) Program Edit
- (13) LCN
- (14) EPG
- (15) Channels Lock
- (16) Parental Lock
- (17) AD
- (18) Info Banner
- (19) Aspect Ratio
- (20) update channel
- (21) Date、Time
- (22) Audio language
- (23) Switch Channel
- (24) FCC/PIP

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7. CBS

(1) DVB Function

- (1) DVB-T/T2
- (2) DVB-C
- (3) DVB-S/S2
- (4) Programs Manage
- (5) EPG/Infor Bar
- (6) Video/Audio
- (7) PVR
- (8) T-Shift
- (9) Subtitle
- (10) Parental Lock

(2) IPTV Function

- (1) UDP TS stream
- (2) UDP MP4 stream
- (3) UDP DVB Subtitle
- (4) UDP Audio
- (5) UDP CAS
- (6) UDP Start/Stop
- (7) UDP HDMI plug
- (8) UDP Unicast support
- (9) UDP Standby wake-up
- (10) UDP channel name display(URL)
- (11) UDP Network disconnection recovery

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1.2. Chip Information

S905X4/S905C2/S905C2L/S905Y4/S905C3/S805X2 :

We choose S905X4/S905Y4 reference platform as our major developing and testing platforms for this release.

Item	S905X4/S905C2/S905C2L	S805X2
CPU	Quad-core ARM Cortex-A55	Quad-core ARM Cortex-A35
GPU	ARM Mali-G31 MP2	ARM Mali-G31 MP2
Memory	32-bit DDR3/4/3L, LPDDR3/4	32-bit DDR3/4/3L, LPDDR4
Video decoding	AV1/H.265/VP9/AVS2-P2 to 4k@60fps	AV1/H.265/VP9/AVS2-P2 to 1080p@60fps
Video Encoding	H.265/JPEG to 1080p@60fps	SW encoder
HDMI-TX	4k60hz	1080p60hz
AV output	CVBS	CVBS
Ethernet	10M/100M/1000M	10M/100M
IP License	Dolby Audio/DTS/Dolby Vision	Dolby Audio/DTS
Demux	Support	Support

Item	S905Y4/S905C3
CPU	Quad-core ARM Cortex-A35
GPU	ARM Mali-G31 MP2
Memory	32-bit DDR3/4/3L, LPDDR4
Video decoding	AV1/H.265/VP9/AVS2-P2 to 4k@60fps
Video Encoding	SW encoder
HDMI-TX	4k60hz
AV output	CVBS
Ethernet	10M/100M
IP License	Dolby Audio/DTS/Dolby Vision
Demux	Support

1.3. Reference Platform Specification

Mbox:

- **Ohm(S905X4)**
EMMC,WIFI QCA6174 ,DDR4 2GB
- **Ohmcas(S905C2)**
EMMC,WIFI QCA6174 ,DDR4 2GB
- **Ohmcas2(S905C2L)**
EMMC,WIFI QCA6174 ,DDR4 2GB
- **Oppen(S905Y4)**
EMMC,WIFI QCA6174 ,DDR4 2GB
- **Oppencas(S905C3)**
EMMC,WIFI QCA6174 ,DDR4 2GB
- **Planck(S805X2)**
EMMC,WIFI QCA6174 ,DDR4 1GB
- **Planck_gtv(S805X2)**
EMMC,WIFI QCA6174 ,DDR4 1.5GB

2. Amlogic Android SDK

2.1. Downloading the Source code

You can download Android S source code by running the following repo commands:

ODM or OEM

Note: Permission for downloading code

1. Google server(Whole code that can build the final firmware, Kernel/uboot are the binary)
2. Amlogic Openlinux server(Kernel/uboot source code)

1. Use below command to get the code in google server(apply the license from Google):

```
$cd ~/<your-google-repo-dir>/
$repo init -u
https://partner-android.googlesource.com/platform/vendor/pdk/ohm/ohm-userdebug/manifest
-b builds/s-tv-amlogic-hybrid-release/STTC.230104.002
$repo sync
```

2. Use below command to get the code in amlogic server(apply the license from Amlogic):

```
$cd ~/<your-amlogic-repo-dir>/
$repo init -u ssh://git@openlinux.amlogic.com/s-amlogic/platform/manifest.git -b s-amlogic -m
openlinux/ott/amlogic/s-amlogic-source-20230106-gtvs.xml
$repo sync
```

Notice: Oversea customers use below command:

```
$cd ~/<your-amlogic-repo-dir>/
$repo init -u ssh://git@openlinux2.amlogic.com/s-amlogic/platform/manifest.git -b s-amlogic
-m openlinux/ott/amlogic/s-amlogic-source-20230106-gtvs.xml
$repo sync
```

Needed storage space: need about 200G storage space.

2.2. How to integrate the code

Some customers need to build the UBoot/kernel source code. Below is the SOP for code integration.

1. Use below commands to get the script tools and patches code from Amlogic.

```
$cd ~/<your-google-repo-dir>/
$git clone ssh://git@openlinux.amlogic.com/s-amlogic/vendor/amlogic/tools/DroidlogicPatch
$cd DroidlogicPatch
$git checkout -t remotes/origin/openlinux/ott/s-amlogic-base-ab4
```

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You Should deploy your source code as follows,

- sdk-root-dir
 - google-repo-dir (5.2 ab4)
 - patch-repo-dir
 - apply_patches.sh
 - cp_src.sh
 - src_conf.txt
 - amlogic-repo-dir (Aml openlinux sdk)

2.Integrate Amlogic proprietary code into Google code. You can simply do this by the tool cp_src.sh got from above.

```
$cd ~/<your-patch-repo-dir>/  
$./cp_src.sh src_conf.txt <your-amlogic-repo-dir> <your-google-repo-dir>
```

3.Apply patches

```
$cd ~/<your-google-repo-dir>/  
$./<your-patch-repo-dir>/apply_patches.sh ./<your-patch-repo-dir>/20230415/patch-list.txt
```

If need to integrate all patches in the patch repo dir, pls run the following cmd :

```
$ ./apply_patches.sh -a .
```

4.Before compilation, you have to do below step.

```
$cd <your-google-repo-dir>/build/make;git checkout -f
```

2.3. Building the Source code

Before build the code, you should execute this command at the code root directory for merge the 2023 02 03 security patches.

```
./vendor/amlogic/common/security_patch/detect.sh
```

Build Bootloader

1. ohm_hybrid(S905X4)

```
cd bootloader/uboot-repo/  
./mk sc2_ah212 --avb2 --vab  
cp build/u-boot.bin.signed ../../device/amlogic/ohm/bootloader.img  
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/ohm/upgrade/  
cp build/u-boot.bin.usb.signed ../../device/amlogic/ohm/upgrade/
```

2. ohmcas(S905C2)

```
cd bootloader/uboot-repo/  
./mk sc2_ah232 --avb2 --vab  
cp build/u-boot.bin.signed ../../device/amlogic/ohmcas/bootloader.img  
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/ohmcas/upgrade/  
cp build/u-boot.bin.usb.signed ../../device/amlogic/ohmcas/upgrade/
```

3. ohmcas2(S905C2L)

```
cd bootloader/uboot-repo/  
./mk sc2_ah221 --avb2 --vab  
cp build/u-boot.bin.signed ../../device/amlogic/ohmcas2/bootloader.img  
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/ohmcas2/upgrade/  
cp build/u-boot.bin.usb.signed ../../device/amlogic/ohmcas2/upgrade/
```

4. oppen_hybrid(S905Y4)

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```
cd bootloader/u-boot-repo/  
./mk s4_ap222 --avb2 --vab  
cp build/u-boot.bin.signed ../../device/amlogic/oppen/bootloader.img  
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/oppen/upgrade/  
cp build/u-boot.bin.usb.signed ../../device/amlogic/oppen/upgrade/
```

5. oppencas(S905C3)

```
cd bootloader/u-boot-repo/  
./mk s4_ap232 --avb2 --vab  
cp build/u-boot.bin.signed ../../device/amlogic/oppencas/bootloader.img  
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/oppencas/upgrade/  
cp build/u-boot.bin.usb.signed ../../device/amlogic/oppencas/upgrade/
```

6. planck/planck_gtv(S805X2)

```
cd bootloader/u-boot-repo/  
./mk s4_aq222 --avb2 --vab  
cp build/u-boot.bin.signed ../../device/amlogic/planck/bootloader.img  
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/planck/upgrade/  
cp build/u-boot.bin.usb.signed ../../device/amlogic/planck/upgrade/
```

Noticed: Bootloader disabled fastboot unlock function by default. Android VTS/GSI or debug need enable fastboot unlock function through --fastboot-write. Like ./mk s4_ap222 --avb2 --vab --fastboot-write.

Build Kernel

1. ohm_hybrid(S905X4)

```
./device/amlogic/common/kernelbuild/mk ohm -v 5.4 -t user
```

2. ohmcas(S905C2)

```
./device/amlogic/common/kernelbuild/mk ohmcas -v 5.4 -t user
```

3. ohmcas2(S905C2L)

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```
./device/amlogic/common/kernelbuild/mk ohmcas2 -v 5.4 -t user
```

4. oppen_hybrid(S905Y4)

```
./device/amlogic/common/kernelbuild/mk oppen -v 5.4 -t user
```

5. oppencas(S905C3)

```
./device/amlogic/common/kernelbuild/mk oppencas -v 5.4 -t user
```

6. planck_gtv(S805X2)

```
./device/amlogic/common/kernelbuild/mk planck -v 5.4 -t user
```

7. planck(S805X2)

```
export KERNEL_A32_SUPPORT=true  
./device/amlogic/common/kernelbuild/mk planck -v 5.4 -t user
```

Build Android

1. ohm_hybrid(S905X4)

```
. build/envsetup.sh  
lunch ohm_hybrid-user  
make -j8
```

2. ohmcas_gtv(S905C2)

```
. build/envsetup.sh  
lunch ohmcas_gtv-user  
make -j8
```

3. ohmcas2_gtv(S905C2L)

```
. build/envsetup.sh  
lunch ohmcas2_gtv-user  
make -j8
```

4. oppen_hybrid(S905Y4)

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```
. build/envsetup.sh
lunch oppen_hybrid-user
make -j8
```

5. oppencas_gtv(S905C3)

```
. build/envsetup.sh
lunch oppencas_gtv-user
make -j8
```

6. planck_gtv(S805X2)

```
. build/envsetup.sh
lunch planck_gtv-user
make -j8
```

7. planck(S805X2)

```
. build/envsetup.sh
export KERNEL_A32_SUPPORT=true
lunch planck-user
make -j8
```

Noticed: Android S will start to use kernel 5.4 and we setup a new kernel and modules build way named standalone build.

Standalone build is a standard kernel and modules build way. We need separate Android and Kernel build. Android S has two UI: Waston UI(Android TV) and Amati UI(Google TV).

Lunch project name	Type
ohm	OTT ATV
ohm_gtv	OTT GTV
ohm_hybrid	DVB GTV
ohmcas	DVB ATV
ohmcas_gtv	DVB GTV
oppen	OTT ATV
oppen_gtv	OTT GTV
oppen_hybrid	DVB GTV
oppencas	DVB ATV
oppencas_gtv	DVB GTV
planck	OTT ATV
planck_gtv	OTT GTV

2.4. Upgrading the firmware

There are 3 ways for update.

- Upgrade with USB burn tool
- Upgrade with OTA
- Upgrade with fastboot

2.4.1. all partitions

If you like to use our usb burn tool to update the images, it's the same as before. We have a fastboot flash-all script to update image, just like Android S. You need not care about the dynamic partitions.

Partition name	make command	imgae name	burning ways
all partitions	make otapackage	1.ohm-ota-xxxxxxx.zip 2.aml_upgrade_package.img 3.ohm-fastboot-xxxxxxx.zip	1.by OTA upgrade 2.by USB burning 3.by fastboot
	make	aml_upgrade_package.img	by USB burning

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2.4.2. physical partitions

In Android S, physical partitions is **bootloader, boot, logo, vendor_boot** etc.

Fastboot should be in **bootloader** mode when flash these physical partitions.

Below is the way of burning each partitions:

Partition name	make command	image name	burning ways
boot	make bootimage	boot.img	fastboot flash boot boot.img
logo	make logoimg	logo.img	fastboot flash logo logo.img
uboot	./mk sc2_ah212 --avb2 --vab	u-boot.bin.signed	fastboot flash bootloader u-boot.bin.signed
odm_ext	make odm_ext_image	odm_ext.img	fastboot flash odm_ext odm_ext.img
vendor_boot	make vendorbootimage	vendor_boot.img	fastboot flash vendor_boot vendor_boot.img

Notes: execute 'fastboot flashing unlock' to unlock the device if wants to use fastboot commands.

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2.4.3. logical partitions

In Android S, logical partitions is **system, system_ext, vendor, odm, product**.

Fastboot should be in fastbootd mode when flash these logical partitions.

Below is the way of burning each partitions:

Partition name	make command	image name	burning ways
system	make systemimage	system.img	fastboot flash system system.img
vendor	make vendorimage	vendor.img	fastboot flash vendor vendor.img
odm	make odmimage	odm.img	fastboot flash odm odm.img
product	make productimage	product.img	fastboot flash product product.img
system_ext	make systemextimage	system_ext.img	fastboot flash system_ext system_ext.img

Notes: execute 'fastboot flashing unlock' to unlock the device if wants to use fastboot commands.

3. Test Report

3.1. Auto Reboot in Normal Temperature

Test Platform	Test Case	Total test time	Test Result
AP222#200	Auto reboot	72H	Pass
AP222#224	Auto reboot	72H	Pass
AP222#228	Auto reboot	72H	Pass
AP222#231	Auto reboot	72H	Pass
AP222#200	uboot autoreboot	72H	Pass
AP222#224	uboot autoreboot	72H	Pass
AP222#228	uboot autoreboot	72H	Pass
AP222#231	uboot autoreboot	72H	Pass

3.2. Auto Shutdown/Suspend in Normal Temperature

Test Platform	Test Case	Total test time	Test Result
AP222#200	Auto shutdown	72H	Pass
AP222#228	Auto shutdown	72H	Pass
AP222#224	Auto shutdown	72H	Pass
AP222#231	Auto shutdown	72H	Pass
AP222#146	Auto suspend	72H	Pass
AP222#164	Auto suspend	72H	Pass
AP222#313	Auto suspend	72H	Pass
AP222#319	Auto suspend	72H	Pass

3.3. 3D Game burning Test Report

Test Platform	Test Case	Total test time	Test Result
AP222#146	3D Game	72H	Pass
AP222#164	3D Game	72H	Pass
AP222#313	3D Game	72H	Pass
AP222#319	3D Game	72H	Pass

3.4. Monkey test Test Report

Test Platform	Test Case	Total test time	Test Result	Comments
AP222#200	Monkey test/300ms	72H	Pass	
AP222#224	Monkey test/300ms	72H	Pass	
AP222#228	Monkey test/300ms	72H	Pass	
AP222#231	Monkey test/300ms	72H	Pass	

Monkey Black list

com.droidlogic.PPPoE
com.droidlogic.miracast
com.droidlogic.mediacenter
com.droidlogic.otaupgrade

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com.droidlogic.tvinput
com.hpplay.happyplay.aw
com.android.tv.settings
com.droidlogic.tv.settings
com.google.android.tungsten.setupwraith
com.google.android.gms
com.android.vending
com.google.android.youtube.tv
com.google.android.videos
com.google.android.tv.remote.service
com.google.android.tvrecommendations
com.google.android.gms.persistent
com.google.android.youtube.tv.recommendations
com.google.android.tv.frameworkpackagestubs
com.google.android.tvcom.droidlogic.videoplayer

3.5. FactoryReset Test Report

Test Platform	Test Case	Total test time	Test Result
AP222#146	Factory reset	72H	Pass
AP222#164	Factory reset	72H	Pass
AP222#313	Factory reset	72H	Pass
AP222#319	Factory reset	72H	Pass

3.6. A/V Burn Test Report

Test Platform	Test Case	Total test time	Test Result
AP222#146	4K H265	72H	Pass
AP222#164	4K H265	72H	Pass
AP222#313	4K H265	72H	Pass
AP222#319	4K H265	72H	Pass
AP222#200	4K H265	72H	Pass
AP222#224	4K H264	72H	Pass
AP222#228	4K H264	72H	Pass
AP222#231	4K H264	72H	Pass
AP222#146	4K VP9	72H	Pass
AP222#264	4K VP9	72H	Pass
AP222#313	4K VP9	72H	Pass
AP222#319	4K AV1	72H	Pass
AP222#200	4K AV1	72H	Pass
AP222#224	4K AV1	72H	Pass
AP222#234	4K H265	72H	Pass
AP222#239	4K H264	72H	Pass
AP222#200	4K VP9	72H	Pass
AP222#146	Youtube	12H	Pass
AP222#164	Youtube	12H	Pass
AP222#313	Youtube	12H	Pass
AP222#319	Youtube	12H	Pass
AP222#188	full format	72H	Pass
AP222#159	full format	72H	Pass
AP222#169	full format	72H	Pass

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3.7. DVB Stress Test

Test Platform	Test Case	Total test time	Test Result	Comments
AH212#637	Auto Search	48H	Pass	
AH212#640	Auto Search	48H	Pass	
AH212#123	Auto Search	48H	Pass	
AH212#637	Single freq. Channel Change	72H	Pass	
AH212#640	Single freq. Channel Change	72H	Pass	
AH212#123	Single freq. Channel Change	72H	Pass	
AH212#637	Multi freq. Channel Change	72H	Pass	
AH212#640	Multi freq. Channel Change	24H	Pass	
AH212#123	Multi freq. Channel Change	72H	Pass	
AH212#637	Subtitle Longterm Playing	72H	Pass	
AH212#640	Subtitle Longterm Playing	72H	Pass	
AH212#123	Subtitle Longterm Playing	72H	Pass	

3.8. Video Format Test Report

Noticed: Please confirm the license before opening the multimedia format.

Extension	Codec Detail	Tested Resolution	Ohm/Ohmcas	Oppen/Oppencas	Planck
.3g2	H263	704x576	Support	Support	Support
.3gp	MPEG-4 Visual	640x480	Support	Support	Support
	MPEG-4	320x240	Support	Support	Support
	H263	320x276	Support	Support	Support
	H264	1920x1080	Support	Support	Support
.avi	AVC	1920x1080	Support	Support	Support
	M-JPEG	1024x576	Support	Support	Support
	RealMagic	720x480	Support	Support	Support
	MPEG-4	720x576	Support	Support	Support
	h264	1920x1080	Support	Support	Support
	FF mpeg MPEG4	640x480	Support	Support	Support
.dat	MPEG-1	352x288	Support	Support	Support
.evo	VC-1	1920x1080	Support	Support	Support
.f4v	AVC(H264)	1280x720	Support	Support	Support
.flv	Sorenson Spark	320x276	Support	Support	Support
	VP6	800x342	Support	Support	Support
.mp4	AVC(H264)	1920x1080	Support	Support	Support
	HEVC(H265)	1920x1080	Support	Support	Support
	4K HEVC(4K H265)	3840x2160	Support	Support	No support
	8K HEVC(8K H265)	8192x4320	Support	Support	No support

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	AV1	1920x1080	Support	Support	Support
	4K AV1	3840x2160	Support	Support	No support
	H263	176x144	Support	Support	Support
.m2ts	AVC	1920x1080	Support	Support	Support
	VC-1	1920x1080	Support	Support	Support
.m2v	MPEG-2	480x576	Support	Support	Support
.m4v	AVC	1280x720	Support	Support	Support
.mkv	WMV3	1280x720	Support	Support	Support
	MPEG-4 Visual	1920x1080	Support	Support	Support
	AVC	1920x1080	Support	Support	Support
	4K HEVC(4K H265)	3840x2160	Support	Support	No support
	8K HEVC(8K H265)	8192x4320	Support	Support	No support
	VP8	1920x1080	Support	Support	Support
	VP9	1920x1080	Support	Support	Support
.mov	MPEG-4 Visual	1280x720	Support	Support	Support
	mjpa	640x480	Support	Support	Support
	H263	320x240	Support	Support	Support
	M-JPEG	640x480	Support	Support	Support
	AVC(H264)	1920x1080	Support	Support	Support
	4K H264	3840x2160	Support	Support	No support
.mpeg	MPEG-2	1920x1080	Support	Support	Support
	VC-1	1920x1080	Support	Support	Support
	MPEG-1	720x576	Support	Support	Support
.mts	AVC	1440x1080	Support	Support	Support
.ogm	XVID	640x480	Support	Support	Support
.PMP	H264	480x272	Support	Support	Support
.tp	MPEG-2	1920x1088	Support	Support	Support
.ts	HEVC(H265)	1920x1080	Support	Support	Support
	4K HEVC(4K H265)	3840x2160	Support	Support	No support
	8K HEVC(8K H265)	8192x4320	Support	Support	No support
	MPEG-1	1920x1080	Support	Support	Support
	avs	720x576	Support	Support	Support
	avs+	1920x1080	Support	Support	Support
	avs2	3840x2160	Support	Support	No support
	AVC	1920x1080	Support	Support	Support
	MPEG-2	1920x1080	Support	Support	Support
.vob	MPEG-2	720x576	Support	Support	Support
.wmv	WMV1	640x480	Support	Support	Support
	WMV2	768x432	Support	Support	Support
	WMV3	1920x1080	Support	Support	Support
.webm	VP8	1920x1080	Support	Support	Support
	VP9	1920x1080	Support	Support	Support
	4K VP9	3840x2160	Support	Support	No support
	8K VP9	8192x4320	Support	Support	No support
	AV1	1920x1080	Support	Support	Support
4K	AVC(H264)	3840x2160	Support	Support	No support
	HEVC(H265)	3840x2160	Support	Support	No support
	VP9	3840x2160	Support	Support	No support

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	AV1	3840x2160	Support	Support	No support
8K	HEVC(H265)	8192x4320	Support	Support	No support
	VP9	8192x4320	Support	Support	No support
	AV1	7680x4320	Support	Support	No support

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4. License version

Licensed Feature Name	Version	Document version
Dolby Digital Plus for Android 32bit	v20230104	In firmware Readme text
Dolby Digital for Android 32bit	v20200723	In firmware Readme text
Dolby MS12 V1.3.2 for android 32bit	v20230321	In firmware Readme text
Dolby MS12 V2.4 for android 32bit	v20220805	Amlogic_Dolby_MS12_v2_release_note_V0.5
DTS-HD for android 32bit	v20230103	In firmware Readme text
DRA Decoder for android	v20210728	In firmware Readme text
Secure OS TDK	TDK is included in the code	For S905X4/S905C2/S905C2L S905X4 Secure Key Provision User Guide (3.4) S905X4 Provision Efuse Object User Guide (1.1) S905X4 Series TDK Integration User Guide(3.5) For S905Y4/S905C3/S805X2 S905Y4 Secure Key Provision User Guide (3.4) S905Y4 Provision Efuse Object User Guide (1.1) S905Y4 Series TDK Integration User Guide(3.5)
Widevine	L1/L3	Amlogic WideVine Integration User Guide v0.9
Playready		Amlogic PlayReady Integrate User Guide v1.40
Dolby Vision 2.6 for Android kernel5.4.210 64bit	Ohm kernel5.4.210 64bit dovi.ko stb_release_20230215 dovi_fw_u2019.bin release-20210413 Oppen kernel5.4.210 64bit dovi.ko stb_release_20230215 dovi_fw_u2019.bin release-20210413	Amlogic DolbyVision Release (v0.5)
HDCP 1.4 for HDMI TX for Android	v20180408	Amlogic_HDCP1.4_Tx_Release_Note_V0.1
HDCP 2.X for HDMI TX for Android	v20230225	Amlogic_HDCP2.2_Tx_Release_Note_V1.1
Verimatrix	vendor/vmx/bootloader(m9x4/openlin ux-android-s: ccf71ab76bcd6aa34d2e8b7fd359ce 612db1904) vendor/vmx/bootloader(m9y4/openlin ux-android-s: 11ddf9b5b8ccf9dff041a20a911218c 810dc8b1) vendor/amlogic/prebuild/verimatrix(r-a mlogic: 440397cc332823c4923d4d9e2bd1c5 08fb9f790a) vmx/sdk-rel(m9x4-rel: 7dbf941542b19e359dc63e9b5349c0 56e82507ab) vmx/sdk-rel(m9y4-rel: ea3f467a4f2c2d82b3449f6242e1346 18e1ad8f7)	OHM S905X4 Verimatrix Android SDK Integration User Guide (2.9) S905X4 Verimatrix Android Client Integration User Guide (5.3) Amlogic CAS - Verimatrix Adaption API Guide-v1.0 Oppen S905Y4 Verimatrix Android Client Integration User Guide (8.1) S905Y4 Verimatrix Android SDK Integration User Guide (1.8) Amlogic CAS - Verimatrix Adaption API Guide-v1.0

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	vendor/vmx/vmx-rel(master: a0fbf908c1a3f8019dae101395f217df 651f659a)	S905C3 Verimatrix Android SDK Integration User Guide (1.2) Planck S805X2 Verimatrix Android Client Integration User Guide (8.3)
Nagra TKL/NOCS	nagra-sdk(projects/openlinux/v2.2: e3c75a7e8f87027fb371be4c00527e5 6b446bc07) nagra-cashal(projects/openlinux/v3.0: ea492c3b3c4ad26227bbb65d47541f e659f35eb3) nagra-sdk(projects/openlinux/v3.4: f61f99386bac8c7ef0c251fbbe3f9215 559a0877)	For S905X4 S905X4 Nagra TKL Android MediaCasV2 Integration User Guide (1.2) S905X4 Nagra TKL Android SDK Integration User Guide (1.4) For S905C2 S905C2 Nagra NOCS Android MediaCasV2 Integration User Guide (1.2) S905C2 Nagra NOCS Android SDK Integration User Guide (1.4)
Synamedia		For S905C2 Amlogic CAS - S905C2 Synamedia SDK Integration User Guide (1.6) S905C2 Synamedia Android SDK User Guide (1.10) S905C2 Synamedia Bootloader Design (1.2)

5. License path

Licensed Files	Path
HDCP TX firmware.le	/odm/etc/firmware/firmware.le
Dolby	/oem/lib/libHwAudio_dcvdec.so
MS12 V2	/oem/lib/ms12/libdolbyps12.so
MS12 V1	/oem/lib/libdolbyps12.so
Dolby vision	/oem/overlay/dovi.ko /oem/firmware/dovi_fw.bin

6. Reference

6.1. Documents

Functional	Documents
Device Vendor SCS Secure boot v3	Amlogic S905X4 Series Secure Boot User Guide Amlogic S905Y4 Series Secure Boot User Guide Amlogic Linux SCS Signing Tool User Guide
Secure Mode	Security Mode Configuration Guide Production Mode Configuration Guide
Provision key	S905X4 Secure Key Provision User Guide Revision S905Y4 Secure Key Provision User Guide Revision
TEE Development Kit (TDK)	S905X4 Series TDK Integration User Guide S905Y4 Series TDK Integration User Guide
AVB2	Amlogic Android AVB2 and Keymaster User Guide

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6.2. Tools

Tool name	version
USB burning tool	Aml_Burn_Tool_V3.2.8
Customization	Amlogic customization V.2.1.3
Linux Secure boot V3	Aml_Linux_SCS_SignTool V2.4.0
adb	1.0.41
fastboot	31.0.3

You can download latest SDK Platform-Tools from

<https://developer.android.com/studio/releases/platform-tools.html#downloads>

6.3. *Dynamic fingerprint*

Although Android TV devices are similar, their system properties can differ. To help reduce the build load, the Android TV OEM partition has been released for use with Android TV on Android 12.

1. Change device/amlogic/oppen/oem/oem.prop
 . build/envsetup.sh
 lunch oppen-user
 make custom_images -j8
 2. copy out/target/product/oppen/oem.img to device/amlogic/oppen/oem/
- As ohm.

6.4. Config DDP MS12 DTS Dolby Vision and HDCP

Dolby Digital Plus, Dolby ms12, DTS or Dolby vision is stored in the oem partition.

1. Copy libHwAudio_dcvdec.so to vendor/amlogic/common/prebuilt/libstagefrighthw/lib/libHwAudio_dcvdec.so
2. Copy libdolbyps12.so to device/amlogic/common/dolby_ms12/install/encrypted_lib/libdolbyps12.so
3. Copy libdolbyps12.so(ms12v1) to device/amlogic/common/dolby_ms12/install/encrypted_lib/ms12v1/libdolbyps12.so
4. Copy libHwAudio_dtshd.so to vendor/amlogic/common/prebuilt/libstagefrighthw/lib/libHwAudio_dtshd.so
5. Copy dovi.ko to device/amlogic/ohm/files/dovi.ko , as ohmcas/oppen etc.
6. Copy dovi_fw.bin to device/amlogic/ohm/dovi_fw.bin , as ohmcas/oppen etc.
7. Copy hdcp_tx22 and firmware.le to vendor/amlogic/restricted_libs/hdcp_tx22/

Enable Dolby Digital Plus, Dolby ms12, DTS or Dolby vision Macro.

```
TARGET_BUILD_WITH_DOVI :=true
TARGET_DOLBY_VERSION := ms12_v2    or set ms12_v1 ddp_only
TARGET_DTS_VERSION :=dtshd
```

Generate OEM partition

```
. build/envsetup.sh
lunch ohm_hybrid-user
make custom_images -j8

copy out/target/product/ohm/oem.img to device/amlogic/ohm/oem/oem_hybrid/

As oppen etc.
```

6.5. Configuration CBS

You can download Android S CBS source code by running the following commands (You need apply license from Google)

```
$cd ~/<your-google-repo-dir>/vendor/  
$mkdir -p google/tv  
$cd ~/<your-google-repo-dir>/google/tv  
$git clone https://partner-android.googlesource.com/platform/vendor/google/tv/broadcaststack  
$cd broadcaststack  
$git checkout -t remotes/origin/s-tv-dev; git reset --hard ef487d3215f068a168e46f4750aad9045dedfb78
```

Build Android

```
source build/envsetup.sh  
lunch ohm_cbs-userdebug  
make -j8  
Or oppen_cbs etc
```

Note:about open version

The open version with built-in demod only supports DVB-C scan; The open version of the external demod (the hardware needs to be modified) supports DVB-T/T2/C.